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Designing Effective Flood Early Warning Systems: A Review of Barriers, Best Practices, and Key Characteristics

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ABSTRACT

Flooding is a disruptive and devastating natural hazard for communities all around the world. To combat the negative effects of flooding, it has become a global priority to implement and research flood early warning systems (FEWS). However, previous research did not comprehensively examine both the technological and social dimensions of FEWS nor the knowledge of their availability and status around the world. To address these gaps, this paper completes a narrative review and synthesizes best practices in FEWS warning and system design, and introduces a “FEWS Around the World” repository which catalogs 3–4 examples per region across six continents. Our analysis shows that while most FEWS include design features outlined in the literature such as timeliness, human capacity, and integration, fewer implement novel components related to impact-based warnings, participatory science, bottom-up approaches, advanced technology, and maintaining a preparedness fund, highlighting gaps and opportunities for improvement. Examples demonstrate how these characteristics manifest in diverse contexts, from community-based systems in Nepal to AI-driven systems like Google’s Flood Hub. By bridging design principles with observed global practices, this paper is designed to aid researchers, practitioners, and communities in people-centered FEWS development, implementation, and operation, improving resilient flood risk management.

1 | Introduction

Today, flood disasters account for about half of all the world’s natural disasters (WMO 2021). Flooding can cause loss of life and damage to property (Acosta-Coll et al. 2018), poor physical and mental health, and harm to the economy and natural ecosystems (Ben-Enukora et al. 2025), among other impacts (see Golding 2022). It is estimated that by the end of the 21st century, flood-related damages could increase, in part, due to climate change (da Fonseca et al. 2022).

To mitigate flood impacts, it has become a priority to implement early warning systems (EWS) in communities all around the world. In 2015, the United Nations Environmental Programme declared access to early warning information a basic human right (Golding 2022). Flood EWS (FEWS) focus on flood hazards and impacts. There is no universal definition of a FEWS, but Sukhwani et al. (2019) state that they “empower the flood prone communities to proactively respond to imminent threats by providing a timely prediction of the scale or magnitude, duration and likely damages of the forthcoming

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flood disaster at a particular location and time” (p. 4). FEWS enhance (or create) community resilience to floods (Cools et al. 2016). To be effective, it should be continuous and ingrained into the community’s functions and activities (Kelman and Glantz 2014), offering an opportunity to create a communal culture of preparedness (Golding 2022). FEWS represent a cost-effective component of flood risk reduction strategies. Their implementation typically incurs lower financial costs while delivering substantial societal benefits that span economic, social, physical, health, mental, and work-related costs (Golding 2022).

A FEWS should be designed for the community, but the technological components often receive the most attention (Baudoin et al. 2014). Technology such as forecast models and water gauges are integral to system operation, but if these are the only tools invested in, the FEWS will become top-down. In such systems, hazard-based and highly technical warnings are disseminated by external entities with limited consideration of local audiences, local vulnerabilities and risks, and potential response capabilities (Basher 2006; Perera et al. 2020). This disconnect can undermine the effectiveness of the FEWS, as end-users (those most affected by impending hazards) are rarely involved in system design or operation (Basher 2006). To address this, social components (e.g., the community) should be integrated into the system and technological components contextualized before the event. This enables the development of a people-centered, bottom-up FEWS that is responsive to local needs and capacities (Kelman and Glantz 2014).

An effective bottom-up FEWS will consider the needs and capabilities of flood communities and their disaster response services, working to ensure they know how and when to respond to an impending flood, and can prepare for and mitigate against one too (Smith et al. 2017). A people-centered FEWS is a type of end-to-end early warning value chain composed of four interrelated steps: (1) Disaster risk and hazard knowledge; (2) observations, monitoring, analysis, and forecasting; (3) warning dissemination and risk communication; and (4) preparedness and response capabilities (Basher 2006; Fisher et al. 2021; Majumdar et al. 2021; Golding 2022; ICFM 2024). Golding (2022) further suggests the ideal early warning system also includes four overarching components, including effective governance and institutional arrangements, a multi-hazard approach, consideration of gender perspectives and cultural diversity, and involvement of the local community (Figure 1). Timely and clear communication between each link is crucial because each has its own

objectives, resources, and constraints when receiving, using, and transferring information (Golding et al. 2019). To receive the full benefit of early warnings, all parts of the early warning value chain must operate effectively (Tan et al. 2022).

The warning value chain model will look different for every community, river basin, or country, depending on the resources and investments available. In fact, most operational FEWS have top-down and bottom-up features. In a developed country like the United States, steps 2 and 3 are reliably top-down, but increasingly, the weather enterprise has prioritized a bottom-up approach to steps 1 and 4 (and even step 3 through social science research). However, in England, despite steps 2 and 3 also being reliably top-down, it has not been as effective in engaging at-risk communities in flood management (steps 3 and 4) (Mehring et al. 2018). FEWS can be entirely bottom-up or community-based, a recommended approach for those living in conflict-ridden or poorly governed countries/regions, so as to avoid system disruptions from insecurity and violence (Tupper and Fearnley 2023). In many developing nations, step 2 is top-down, but they lack the ability to “translate early warning into early action” (Golding 2022, VI). No country has fully achieved all four FEWS pillars, while very few have fully functioning end-to-end FEWS (ICFM 2024). There are many FEWS around the world, and each varies in design and purpose, which makes gauging their effectiveness challenging.

Previous research on FEWS has explored best practices in the flood early warning value chain and has noted their importance to flood disaster risk reduction. However, none has aimed to comprehensively review the knowledge on FEWS availability and status around the world, nor synthesize their benefits, costs, and challenges (Perera et al. 2019). Other studies have evaluated or measured the effectiveness of FEWS through different criteria and approaches (e.g., Parker and Fordham 1996; Sättele et al. 2016), but they do not wholly consider the impact that a rapidly changing world might have on long-term effectiveness. To address these research gaps, we compare FEWS designs around the world using key characteristics of an effective FEWS (Table 1) and complete a narrative review of best practices in warning and system design to aid in their development, implementation, and operation. Additionally, this paper highlights the role of precipitation forecast products, such as the U.S. National Weather Service’s (NWS) Weather Prediction Center’s (WPC) precipitation products, as central elements in a FEWS, important for communication and requiring bottom-up development to adequately address warning information and dissemination needs.



FIGURE 1 | People-centered flood early warning system (adapted from Golding 2022).

TABLE 1 | Summary of key characteristics of effective flood early warning systems.

	Characteristic	Definition	Key example (see FEWS repository)
Adapted from Kelman and Glantz (2014)	Continuity	FEWS should be operationally-continuous and built into the community as a social process.	Cayman Islands Smart FEWDS (Cayman Islands)
	Timeliness	FEWS should provide enough lead time for decision-making.	Early Warning System (Argentina)
	Transparency	FEWS should be transparent in that raw and unprocessed data can be confusing for end-users, but so can withholding information, so keeping a record of relevant information is beneficial for interested stakeholders and to understand where improvements can be made, post-event.	European Flood Awareness System (pan-Europe)
	Integration	FEWS should be effectively integrated into society and understood by end-users as to how it works within larger contexts.	Community-based EWS (Nepal)
	Human capacity	FEWS should employ both human operators and experts.	Harris County Flood Warning System (USA)
	Flexibility/adaptability	FEWS should be able to expand its duties to other hazards and vulnerabilities, as and when needed; further, operators should continually work to improve the FEWS and adapt to changing needs and considerations.	DisasterAWARE (Global)
	Catalysts/patterns	FEWS should have thresholds, a trigger mechanism, and/or schedule (pattern) for disseminating risk information and warnings.	Flood Forecasting and Early Warning System (Belgium)
Proposed extension, based on this review	Impact-based warnings	FEWS should include impact-based warnings that share the hazard as well as risk level and potential impacts, with their spatial and temporal likelihoods, using predefined categories.	Impact-based Flood Early-Warning Trigger System (Uganda)
	Partnerships	FEWS should promote, establish, and maintain long-term partnerships with a diversity of entities and stakeholders, including the community it serves.	FloodNet NYC (USA)
	Preparedness fund	FEWS should establish, secure, and maintain a preparedness fund which can support system development, implementation, and operation costs. The existence of a one-time funding stream(s) does not meet this characteristic; instead, FEWS preparedness funding should aim to support ongoing operation and development and include planning for continued funding as much as possible.	Total Flood Warning System (Australia)
	Governance structure	FEWS should have a good governance structure in place that considers the unique characteristics/situation of the community, designed to prioritize all four steps of an end-to-end early warning system to develop resilience to flooding.	Central America Flash Flood Guidance System (Central America)
	Participatory science	FEWS should include opportunities for its end-users to be more involved with development and operational processes, including contributing or collecting necessary data.	SIATA (Colombia)
	Artificial intelligence and related technology	FEWS should consider integrating artificial intelligence and related advanced technology into its existing structure(s) to enhance and expand on current capabilities.	Google's Flood Hub (Global)
	Bottom-up approach	FEWS should plan, integrate, and maintain its social and technological components for the community it serves with end-users as the central focus.	Community-based FEWS (Hindu Kush Himalayas)

Section 2 describes our methodology for the narrative review and development and analysis of a “FEWS Around the World” repository. Section 3 summarizes best practices and identified barriers in FEWS design and each step in the flood early warning value chain, gleaned from a review of existing literature, providing crucial background to our FEWS review. Section 4 introduces and synthesizes novel key characteristics of FEWS as they relate to system development and operation; information on key characteristics is derived from the literature review, and subsequent case studies of FEWS from the repository interpret the characteristics in practice. Section 5 discusses the contribution and implications of our review, and suggests future research as it relates to our findings. Section 6 shares final conclusions and our next phase of research.

2 | Methodology

This paper adopts a narrative review approach, aiming to synthesize the breadth of research and practice on FEWS. Rather than conducting a statistical meta-analysis, we systematically identified and qualitatively analyzed peer-reviewed and grey literature to capture key characteristics, barriers, and best practices across diverse contexts. Narrative reviews are particularly suited to fields such as disaster risk management, where evidence is heterogeneous and fragmented, and the goal is to provide an overview of concepts, approaches, and gaps rather than quantify effect sizes. Our synthesis highlights common design features and challenges in FEWS and introduces a repository of global examples to illustrate how these characteristics are operationalized in practice.

To inform our review of best practices in warning and FEWS design, we completed a search of the literature using Google Scholar, institutional libraries, and various journal databases (e.g., *IEEE Xplore*, *ScienceDirect*, and *SpringerLink*). We collected 89 papers from 39 different journals as well as several book chapters and numerous reports for the narrative review. Recently published literature (i.e., in the last 10 years), frequently cited work, and reports on flood or early warning systems were given importance. Reputable journals focused on components of the flood early warning value chain were also prioritized, including risk knowledge and hazard preparedness (e.g., *Natural Hazards and Earth System Sciences*), meteorological forecasting and risk communication (e.g., *Bulletin of the American Meteorological Society* and *Weather, Climate, Society*), technological advancements (e.g., journals from the *Institute of Electrical and Electronics Engineers*), and disaster response and management (e.g., *International Journal of Disaster Risk Reduction* and *Journal of Flood Risk Management*). We largely limited our review to peer-reviewed literature or resources from reputable organizations (e.g., entities within the United Nations and NWS). The breadth of literature to review was also limited by the fact that most authors are native English speakers and of US origin; we acknowledge potential literature and analytical biases as they pertain to lived experiences (with flooding and FEWS) and research foci (FEWS best practices and challenges). Lastly, our literature review included studies on both EWS and FEWS; for simplicity, EWS and FEWS will be referred to as FEWS in our narrative review.

Our inclusion search for operational and experimental FEWS in the “FEWS Around the World” repository was first opportunistic (i.e., those we came across during the literature review) and then systematic (using key search words like “flood early warning systems” and “flood early warning systems in [insert country/continent]”) based on identified regional gaps (e.g., South America and Africa) and the availability and sufficiency of information on systems in the literature and online; we prioritized prominent and well-documented systems and then filled regional gaps using available information. We include systems located in Africa, Asia, Europe, North America, Oceania (representing Australia and Other Pacific Island Nations), and South America, as well as EWS that have global coverage, with an aim of reviewing 3 to 4 systems per region. Within each region, we tried to capture a diversity of FEWS in purpose and spatial coverage. Although a selective sample of FEWS is not representative of all that exists, the repository provides valid insights because the literature review of best practices in FEWS design guided its construction, including system inclusion and review. Information on the FEWS is from the literature review (e.g., peer-reviewed literature and reports), Internet search and online resources (e.g., conference abstracts and presentations and news articles), or the system itself (e.g., FAQs, methodology reports, and/or the dashboard) and all data sources are cited in the repository. We collected 34 papers as well as book chapters and numerous reports and webpages for the FEWS review. Due to limitations in information availability and sufficiency, we did not define the kinds of resources to include or exclude but made an effort to consult only the most up-to-date information at that time (February and May 2025). We recognize that our approach to FEWS research and inclusion might bias toward better-documented systems and so welcome contributions of including more FEWS or adding missing or updating information (see the repository for more details).

We completed three phases of research. The first phase focused on best practices of FEWS for flood risk communication to diverse audiences, among other components; key search words included “effective flood risk communication,” “flood early warning systems,” and “people-centered early warning systems.” Finding most literature related to risk communication or technological advancements, we confirmed the need for a review of the entire flood early warning value chain. This conclusion guided the second and third phases of research. The second phase focused on EWS and community resilience to natural hazards; key search words included “resilience” and “resilience in early warning systems”. In the third phase, we explored the World Weather Research Programme’s High Impact Weather (HIWeather) project, a 10-year activity designed to promote and carry out research on, and effectively improve, high-impact weather-related warnings (Golding et al. 2023). We reviewed HIWeather’s scientific publication lists and selected papers based on their title and abstract, looking for words and themes similar to what we had reviewed up to that point. A research question arose from this literature review: how do FEWS differ operationally around the world, and why? This paper aims to answer this question through a “FEWS Around the World” repository. While not comprehensive of all that exist around the world, the purpose of the repository is to catalog the design and function of select FEWS and to inform how they implement best practices and key characteristics for effectiveness.

Considering a FEWS design is dependent on a number of factors that may affect effectiveness, the repository organizes systems based on key characteristics of an EWS described in Kelman and Glantz (2014), and additional recommended key characteristics based on best practices of warning and system design found in the literature, thus providing an adaptable approach to developing, implementing, and operating effective FEWS. These characteristics include: *continuity, timeliness, transparency, integration, human capacity, flexibility, and catalysts/partners, impact-based warnings, partnerships among all FEWS stakeholders, a preparedness fund, governance structures, participatory science, artificial intelligence and related advanced technology*, and be developed, implemented, and operationalized through a *bottom-up* approach. Table 1 defines these characteristics; the first seven definitions are sourced and summarized from Kelman and Glantz (2014), a highly cited chapter in a book on early warning systems for climate change, and the second seven are from a qualitative, iterative thematic synthesis of the reviewed literature to identify additional features cited as important for system effectiveness. Publications were examined for recurring themes related to system design, governance, community engagement, and technological approaches. These emergent themes were grouped into supplementary characteristics to extend the existing framework. The full set of characteristics, along with refined definitions and their operationalization in practice, is presented in Table 1 and described in detail along with our results in the Characterizing FEWS Around the World section.

There are many FEWS around the world and each varies in size/coverage, design and purpose, and the resources and investments available to them, which makes gauging effectiveness challenging. We have focused on a qualitative assessment for this study, and have not developed quantitative metrics to assess or validate the effectiveness of EWS included in the “FEWS Around the World” repository. To systematically organize systems, we applied a binary coding approach: each characteristic was marked as present or absent based on information reported in the literature, reports, or online sources. Coding decisions were guided by definitions of the characteristics from the reviewed literature and refined iteratively during the synthesis process. Ambiguous or unclear cases were discussed among the research team to reach consensus and all were reviewed by at least two research members. This approach provides a qualitative mapping of FEWS features across contexts and allows readers to explore patterns and gaps in system design without assigning a quantitative measure of effectiveness, which is context-dependent (Kelman and Glantz 2014). In this approach, we affirm that effectiveness is dependent on the specifics of the FEWS, and show how including all 14 characteristics—based on reviewed best practices—better equips the system to help protect against flooding impacts.

3 | Best Practices in FEWS Design

3.1 | Disaster Risk and Hazard Knowledge (Step 1)

Disaster risk and hazard knowledge forms the foundation of FEWS (Perera et al. 2019). Flood risk assessments can build this knowledge by informing about land use, population, the

financial value of at-risk assets, and availability of flood hazard maps (Perera et al. 2019), and can trace existing vulnerabilities to determine what caused them and understand why they remain. To effectively mitigate disaster and achieve resilience to flooding, all stakeholders must understand and educate each other on the hazards that will affect a community, their vulnerabilities and exposure (who and what will be at risk/impacted), and how impacts might cascade into and onto others (Golding 2022), using such knowledge to make better decisions (Acosta-Coll et al. 2018). Participation from all stakeholders builds mutual trust, end-user and inter-agency cooperation, and allows communities to shape their FEWS (Parker and Priest 2012). However, achieving participation from all FEWS stakeholders can be challenging; social science research in England has found an effective strategy to better engage at-risk communities in flood management (see Table A1).

Another way to build disaster risk and hazard knowledge is to enhance communal flood literacy—local knowledge about flooding, designed to actively involve those at risk in local flood management. Literacy can be developed through informal science education, which allows community members to learn about flood risk and preventive strategies (Parker and Priest 2012). This education should come from multiple sources, including leaders in the community, and not just scientists and risk managers (Mileti 1995; Baudoin et al. 2014). The best time to integrate FEWS components into the community, including flood literacy, is during periods of calm weather.

3.2 | Observations, Monitoring, Analysis, and Forecasting (Step 2)

The second step of the FEWS value chain includes technology and humans who observe and monitor hydroclimatic data, analyze collected data to support observations and monitoring as well as flood detection. They then inform stakeholders about future or imminent flooding (Perera et al. 2021). A significant barrier to this step and FEWS implementation is that data access, availability, and process management are highly variable (Tan et al. 2022; Hoffmann et al. 2023). Many countries do not have reliable forecast models, sufficient gauging stations, or the technical expertise and human power to adequately forecast precipitation and issue timely warnings (Perera et al. 2019).

National Meteorological and Hydrological Services (NMHS) are valuable to this step because of the information and insight they can provide and so work closely with FEWS operators. For example, Harris County Flood Warning System (HCFWS) shares rainfall and stream/bayou level data with NWS, who might consider that data in tandem with their own quantitative precipitation forecasts (QPFs) and river flood outlooks, to inform timely issuance of official warnings and advisories. Further, WPC maintains both deterministic and probabilistic QPF products that are used throughout the United States to support flood risk assessments and forecasting (WPC 2025). In other instances, NMHS use FEWS to complement or shape their own warnings, like the European Flood Awareness System (EFAS) does for European entities. No matter how FEWS are used, risk communication should be impact-based, and effective observation, monitoring, analysis, and forecasting tools should

be both hazard- and impact-based, and in continuous operation (WMO 2015; Taylor et al. 2018; Zhang et al. 2019).

3.2.1 | Hazard-Based Forecasting

Flood forecasting varies based on the type of event and geographic scale. Most FEWS rely on real-time rainfall information, high-resolution probabilistic (ensemble) numerical weather and flow forecasts, and well-calibrated hydrological model systems; local data and information and skillful forecasters are also beneficial (Merz et al. 2020). Forecasting systems are designed to predict flooding using gauge, radar, and/or projected rainfall data. An example of projected rainfall data is QPFs, which “are predictions of the amount of rainfall over a future time period produced by NWS forecasting tools” (p. 66); the QPF can be useful for a range of flooding types including urban stormwater flooding (Fang et al. 2023). In most cases, a meteorological or flood service will be responsible for this step and work in coordination with relevant authorities to ensure adequate communal flood protection. Rainfall and water level gauges might be automatic, semi-automatic (data recording is automated and collection and transmission is manual), or entirely manual (Perera et al. 2019). Observation and monitoring networks evolve over time in quantity, quality, and diversity due to technological advancement and to meet the needs of stakeholders (Majumdar et al. 2021).

3.2.2 | Impact-Based Forecasting

While hazard-based forecasts share the physical characteristics of flooding, impact-based forecasts include expected impacts by considering quantitative (e.g., estimates of economic or physical losses and damages) and qualitative (e.g., flood likelihood/potential impact severity thresholds) criteria (Merz et al. 2020). Vulnerability and exposure information collected by flood risk assessments can make flood forecasts more impact-based (Anderson-Berry et al. 2018).

Transitioning to impact-based forecasts requires input from end-users and professional risk communicators who have knowledge of local conditions (Merz et al. 2020). Traditional knowledge and expertise can supplement a lack of robust technological access and foster better forecasts/warnings, as studied in Uganda (see Table A1).

3.3 | Risk Communication and Warning Dissemination (Step 3)

Communication and dissemination are distinct: dissemination refers to how risk information or the warning reaches end-users, and communication relates to its content (Golding 2022). Both, as integral interrelated components of step 3, must be successful for an effective FEWS. Flood risk messaging should be simple, clear, and useful (Acosta-Coll et al. 2018), understandable for a diverse audience, tailored to their interests, needs, and values (Perera et al. 2020), and informed by accurate, timely, and complete data (Mileti 1995). Kelman and Glantz (2014) suggest that warning information should include the five Ws and one H: who, what, when, where, why, and how. A warning should include

who is at risk, what the likely hazard impacts and/or vulnerabilities of concern are, when impacts are expected, where locations at risk are, and why there is a threat and/or vulnerabilities that exist. Further, a message should articulate who the warning is and is not for, what protective actions they should take (Mileti 1995), and when (da Fonseca et al. 2022).

The message should be communicated through multiple channels (e.g., television, radio, and social media), which should be identified before the flood event and robust enough to avoid potential transmission disruption to reach the most number of people (Mileti 1995; Fisher et al. 2021), and include approaches that overcome technological barriers to receiving messaging such as the use of megaphones and blow horns, loudspeakers (Mileti 1995), flood sirens (Intrieri et al. 2020), and door-to-door warnings (Ben-Enukora et al. 2025).

Technological advancements have changed the way risk is communicated and warnings are disseminated. Smartphones have sped up the time it takes for forecast, hazard, and impact information to move from one person to another and the number of people who receive it. End-users can program weather apps to alert them of warnings in a specific place and can receive a push notification for certain hazards near them. Further, social media platforms provide a place for those at risk or experiencing flooding to relay observations, effectively becoming social sensors (Cecilia et al. 2021) (see Table A1) and to post official warnings from trusted sources (Anderson-Berry et al. 2018).

Even when a warning message is received, people still might not act on it because they lack risk awareness, dismiss messaging (i.e., degree of trust) (Perera et al. 2020), or misunderstand or cannot read it (i.e., language barrier) (Parker and Priest 2012). Further, costs always have a negative impact on promoting risk reduction behaviors, and society protects itself from extremes to the extent that it is affordable and readily manageable (Golding et al. 2023). In other instances, end-users may not respond because of emotional or cultural attachment to their property (Maidl and Buchecker 2015), or believe hard defenses like flood walls will protect them (Rollason et al. 2018) or that they are not at risk (Mileti 1995). These barriers highlight that flood risk communication should focus on risk severity, impacts, and benefits of risk reduction measures as opposed to hazard severity, costs, and perceptions of vulnerability (Weyrich, Mondino, et al. 2020).

3.4 | Preparedness and Response Capabilities (Step 4)

A people-centered FEWS enables action, ensuring those at risk are capable of responding to early warning information before an event and are prepared to respond afterward (Golding 2022). A noted challenge is that people often shift protective action responsibilities to authorities; for a FEWS to be successful, risk reduction strategies must be shared between citizens and authorities. Civil society organizations are helpful in disaster response and risk reduction, as they often have a better understanding of local communities and thus can use tailored communication, knowledge, and practices to break down barriers (Perera et al. 2020). Community-based organizations (CBOs) localize flood risk knowledge, develop social networks, and link flood

risk management strategies with people's livelihoods to improve the creation and dissemination of flood risk communication, protective activities before flood events, and the overall standard of living in vulnerable communities (Li and Lin 2023). CBOs work to improve communication and relationships between a community and their authorities, educate communities about flood risk, and invest in physical preparedness and response materials (Li and Lin 2023). It has been found that volunteers from these kinds of groups can often increase risk perceptions better than government officials (Ben-Enukora et al. 2025). Importantly, preparedness and response capabilities are as unique as the people adopting them; Balay-As et al. (2018) study how indigenous knowledge can be better integrated into FEWS (see Table A1).

3.5 | Bringing It Together

There are operational FEWS that have worked to implement best practices of each step in their design. One such example is the community-based flood early warning system (CBFEWS) located in the Lower Karnali River Basin in Nepal. End-users in the basin contend with poor disaster preparedness and risk management, while forecasting river flows remains a challenge, especially during the monsoon season (Smith et al. 2017). The

international non-governmental organization Practical Action has been working with the Nepal Department of Hydrology and Meteorology (DHM) to establish and keep this CBFEWS operational, which includes trained local gauge readers to monitor water levels, multiple communication avenues, and community education (see Table 2). Practical Action assessed local communities and national government representatives involved with the CBFEWS, finding that 93% of respondents believe the early warning information is reliable. Further, 83% now understand the messages because trusted local leaders and others are involved with the system, and volunteer networks disseminate messages in local languages (Practical Action 2023). Rai et al. (2020) also surveyed households, finding that the CBFEWS provides enough early warning to protect livelihoods, saving USD 1083 per household per flood event; these savings can be increased by 1.83 times with an additional hour of lead time and 2.56 times with two additional hours.

Designing FEWS with all four steps in mind supports effective systems by focusing on what information is required by end-users and how it should be delivered. This "first mile" paradigm involves end-users immediately at the development stage of a FEWS, as opposed to them being an afterthought at the end of the design stage, which is aptly called the "last mile" paradigm

TABLE 2 | Lower Karnali River Basin community-based flood early warning system.

Step 1: Disaster risk and hazard knowledge	To ensure disaster risk and hazard knowledge, community members and officials have worked to create understandable water-level severity categories, and identified safe locations and evacuation routes.
Step 2: Observations, monitoring, analysis, and forecasting	Trained community gauge readers monitor river levels using water gauges, record readings if the gauge is not automated, and disseminate warnings when severity thresholds are met; Community Disaster Management Committees will also monitor and record flood levels, duration and impacts. Automated gauges send data to a publicly-available web-based FEWS managed by DHM, which then inform flood warning bulletins (Smith et al. 2017). Gauge readers will send or receive warnings via mobile phone or SMS, and adhere to communication procedures and protocols (including access to a list of CBFEWS stakeholder contact information), which are updated before each monsoon.
Step 3: Risk communication and warning dissemination	Communication is three-ways: one is bottom-up in which local gauge readers disseminate a flood warning to their community and those downstream, and the other two are top-down due to the automated water gauge network. District-level governments are alerted when severity thresholds are met, which then trigger automatic sirens, and automatic SMS text messages are sent to relevant professional stakeholders who then communicate with security forces who coordinate wider communal dissemination of warnings.
Step 4: Preparedness and response capabilities	To enhance flood preparedness and early warning response capabilities, the Karnali River Basin provides educational information via several modes of communication including involving the community in arts competitions related to flooding. Further, pre-defined response options and stakeholder roles and responsibilities have been established, adequate access to emergency supplies provided, mock drills performed, and response plans integrated at local and national levels. Community members can be trained as "first receivers" of early warning information in which they know how to interpret, re-package, and disseminate warnings to others, and a community is designated as "response capable" once they appropriately respond to a flood event (Smith et al. 2017). To increase warning lead time and address forecasting challenges, the Karnali CBFEWS has since implemented anticipatory threshold warnings, and probabilistic forecasts based on hydrological modeling which provide 5 h (and sometimes up to 8 h) of advanced notice.

(Kelman and Glantz 2014; Golding 2022). Given this context of best practices for FEWS design, the next section describes findings related to our research on operational and experimental FEWS. We suggest additional characteristics gleaned from the literature that FEWS can include and the need for systems to evolve and adapt in response to technological advancements and wicked global problems like climate change.

4 | Characterizing FEWS Around the World

The “FEWS Around the World” repository (Painter et al. 2025; <https://tinyurl.com/FEWSaroundtheworld>) shares just a glimpse of the variety of flood early warning systems that exist, and organizes them by their key characteristics; for EWS that monitor multiple hazards, the repository is an illustrative tool that focuses only on the flood component. The first seven key characteristics of an effective FEWS are outlined in Kelman and Glantz (2014) and supported by the best practices literature review. However, based on this review, we further refine the definition of *transparency* and add *adaptability to flexibility*, suggesting that systems should be adaptable and able to evolve within changing landscapes (e.g., climate change) and incorporate stakeholder feedback as part of an iterative process. The second seven characteristics represent a synthesis of themes found across the literature and are noted by experts for contributing to an effective or successful FEWS (details are in later sections). The approach for organizing systems by their key characteristics was an iterative process. FEWS components that resemble the definition of a key characteristic were organized in that column; for example, information on warning lead time is organized under the Timeliness column and information on system funding is organized under the Preparedness Fund column.

This review suggests that FEWS are widely accepted as an effective disaster risk reduction strategy, but may not always address the social, physical, and environmental complexity of floods and our growing and technologically evolving world. Considering the rise in occurrence and magnitude of floods, and to better address their complexity, we find FEWS development, implementation, and operation should include the following additional seven characteristics: *impact-based warnings*, *partnerships*, *a preparedness fund*, *governance structures*, *participatory science*, *artificial intelligence and related advanced technology*, and *a bottom-up approach* (see Table 1). Further, systems should be adaptable, able to evolve within changing landscapes and incorporate stakeholder feedback as part of an iterative process, adding *adaptability to the flexibility* characteristic.

We further explore these additional characteristics in the following paragraphs: first by summarizing the overall analysis results, then by defining the characteristics and reviewing FEWS in the repository that realize them. Of the FEWS reviewed for the repository, a majority (see Figure 2) include some version of *continuity* (88%), *timeliness* (100%), *transparency* (88%), *integration* (100%), *human capacity* (96%), *flexibility* (83%), and *catalysts/patterns* (100%), but the purpose, design, and/or functions of these characteristics vary. For example, most operate continuously, but this might mean only

the tools used to measure rainfall operate continuously while forecasts are shared on a (sub-)daily timestep. Relatedly, most are timely, but when and by whom forecasts are issued varies; for example, HCFWS in Texas, United States sends an automated alert when water level or rainfall thresholds are met at a gage location; however, official watches and warnings are issued by NWS. Transparency is also variable across systems, whether they provide the technical methodology, data sources, and/or forecast skill, offer a public FAQ document(s), or a combination of this information; EFAS provides particularly detailed user guides and technical documentation. Most FEWS have some sort of human capacity, some with dedicated staff (e.g., HCFWS) and others human volunteers (e.g., CBEWS in Nepal); however, some rely on autonomous sensors. All FEWS have some mechanism, threshold, or trigger for alerts; however, the severity level, timing, and dissemination of the alert depend on the system.

Most of the systems (88%) reviewed have a *governance structure(s)* in place, and all (100%) have some sort of *partnership(s)*; however, most do not yet include a *preparedness fund* (50%), *participatory science* (46%), or *advanced technologies* (33%). Seventy-five percent have some form of *impact-based warnings*, and 79% have a *bottom-up* approach within their system. Only two FEWS reviewed have all 14 of the characteristics outlined (EFAS and Google’s Flood Hub). While most (93%) of the FEWS reviewed have the original characteristics gleaned from the literature (see Figure 2), less (only 67%) have the additional characteristics proposed in this paper, especially for those related to emerging issues like advanced technology and funding.

4.1 | Impact-Based Warnings

Increasingly, warnings are moving from hazard- to impact-based. This change is because impacts can be tangible (e.g., physical) and intangible (e.g., mental), and because vulnerability and exposure, or the way in which risk is expressed, vary (Golding 2022). Impact-based warnings improve warning understanding, risk perceptions, and likelihood to engage in mitigating behaviors, especially during severe weather events (Weyrich, Scolobig, et al. 2020) by communicating risk level and potential impacts, with their spatial and temporal likelihoods, using defined categories (Golding 2022). Historically, impact-based information has been based on a forecaster’s experience with previous events, and its reliability and effectiveness are limited by the changing impacts of flood events today. To remedy this, forecasts and the models have increasingly considered the hazard as well as exposure (the number of people and assets in the area), and vulnerability (the conditions that make a person susceptible to hazard impacts), and how each will evolve over space and time (Golding 2022).

Mitheu et al. (2023) explore the value of impact-based forecasting in their study in the Katakwi District in Uganda. This area is served by a hazard-based Flood Early Warning trigger System (FEWtS) that uses flood hazard forecasts and risk information for early warnings and their trigger thresholds; an Early Action Protocol defines anticipatory actions based on these thresholds. The study included an impact-based FEWtS that integrates

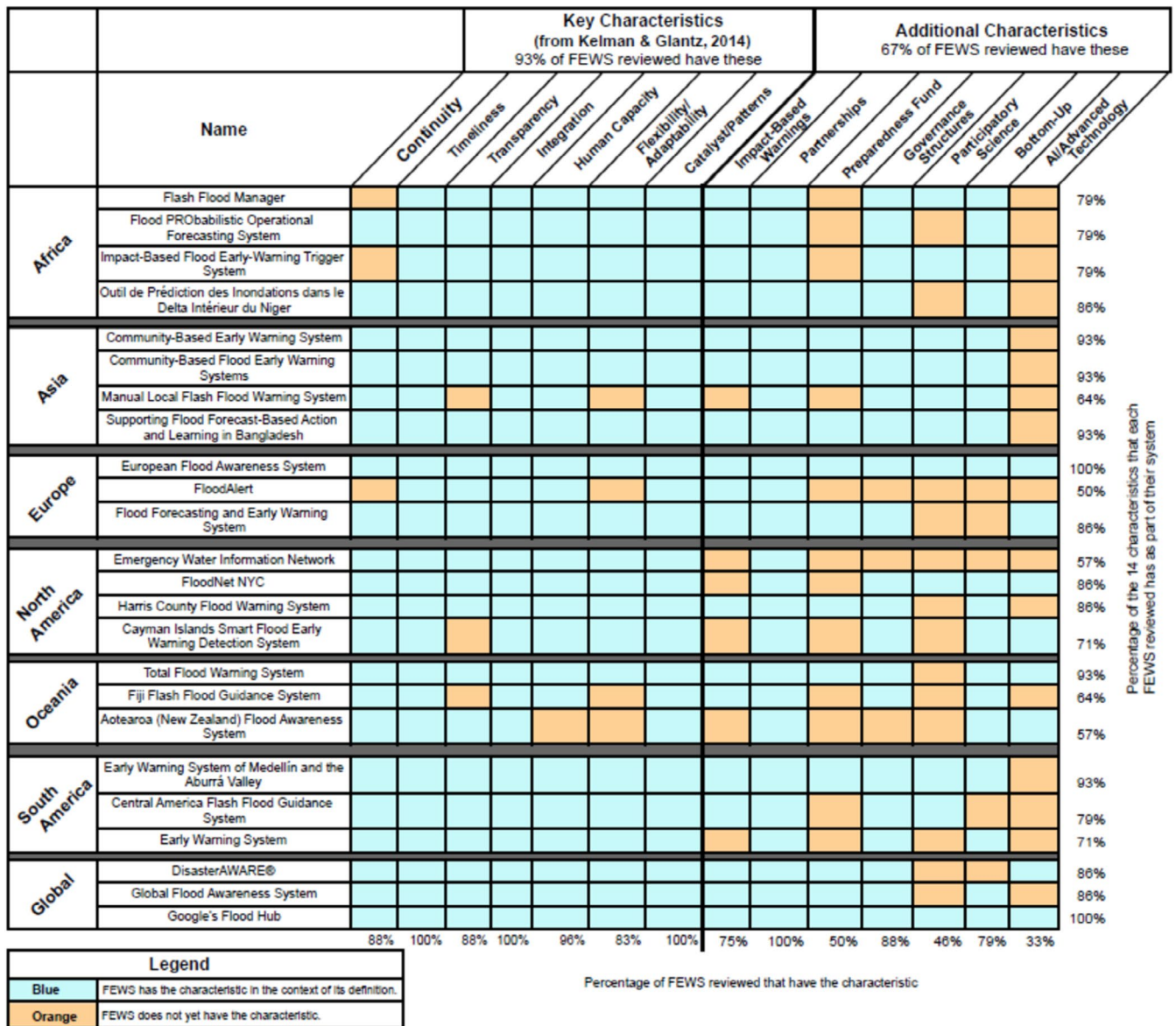


FIGURE 2 | Matrix of FEWS reviewed showing whether they have (blue) or do not have (orange) key characteristics for FEWS design. Percentages of FEWS that have each characteristic are listed horizontally at the bottom and the percentage of all 14 characteristics that each FEWS have are listed vertically on the right. The characteristics from existing literature (Kelman and Glantz 2014) are listed as the first seven columns and the seven new characteristics proposed in this paper are the last seven columns.

livelihood information from local stakeholders and crop cycle databases into the flood hazard forecast to inform variable trigger thresholds and more targeted anticipatory actions, and uses disaster databases to evaluate the forecast performance and compare anticipatory actions with the hazard-based system (Mitheu et al. 2023). The impact-based system performed better in terms of probability of flood detection and missed events while both kinds of systems could benefit from incorporating local contextualized information.

4.2 | Partnerships

An effective FEWS will promote, establish, and maintain partnerships and communication between all stakeholders so that information is not lost in the warning value chain (Golding 2022);

any failed “connection” could render the FEWS ineffective and unsuccessful (Hoffmann et al. 2023). For example, post-event damage surveys typically don't address the total extent of impacts, like those on human health; partnerships between surveyors and local hospitals can provide access to anonymized hospital admissions data (Golding et al. 2019) to fill this information gap. Partnerships between the private and public sector can help developing countries secure timely preparedness and response resources (Hettiarachchi and Weerasinghe 2014). No matter the complexity of the FEWS, it is encouraged that a NMHS, such as the NWS in the United States, or a similar entity, become a partner, as they can provide guidance, data and other resources, technical support, and outreach and education (Fang et al. 2023). Partnership is a key component of FloodNet NYC which fosters and maintains relationships with a diversity of stakeholders, including education and research institutions,

government offices and agencies, and other community groups; these stakeholders assist in choosing the locations of flood monitoring sensors, developing tools and resources for data access and visualization, and enhancing local flood impact knowledge and resilience (Silverman et al. 2022; Mydlarz et al. 2024).

4.3 | Preparedness Fund

Financial investments in FEWS vary around the world and can come from national/local governments, international donations, loans, bi-lateral funding, public-private partnerships, the private sector, and/or national/international research. While developing areas in Asia and Africa receive FEWS investment from almost all of the above sources, developed economies like those in Europe rely on governments, international donations, and loans (Perera et al. 2019). Further, many studies note that certain pillars or FEWS functions are (un)derfunded (Rogers et al. 2019; Golding 2022; ICFM 2024). To address this, a preparedness fund should be established and maintained by the primary FEWS manager, which can support system development and implementation, as well as operational costs like flood risk education activities, updating forecast technology, risk communication social science research, expanding access to warning dissemination, and stockpiling emergency supplies and response resources. Continual and secure funding can mitigate damages from flooding impacts; for example, for every 1 Euro invested in the EFAS, up to 409 Euros in damages can be avoided under certain circumstances (Pappenberger et al. 2015). The International Centre for Integrated Mountain Development (ICIMOD), an intergovernmental research institution, provided seed money to manufacture CBFWEWS instrumentation in Dihiri village in the Dhemaji district of Assam, India, that provided early warning of flooding during a September 2013 event and helped communities save around USD 3300 in assets (Shakya et al. 2019).

4.4 | Governance Structures

A long-lasting people-centered FEWS requires strong political commitments and durable institutional capacities (Basher 2006). However, these are not always available, warranting the need for CBFWEWS. This highlights the difference between *government* and *governance*. An effective FEWS requires good governance, which includes accountability and public engagement (Mysiak et al. 2013). Without strong public engagement, for example, adequate political and budgetary support for FEWS may not be available (Basher 2006), which can limit response and preparedness capabilities and the ability to foster resilience. Monitoring, analysis, and forecasting data should be shared with all levels of governance (local to international) (Tupper and Fearnley 2023); and countries and communities should work together to ensure everyone receives data that can better inform FEWS warnings.

The Central America Flash Flood Guidance System in Central America is an example of good FEWS governance. A part of the larger Flash Flood Guidance System with Global Coverage project, activities are governed by a Memorandum of Understanding, and the system includes two Regional Centers (located in Costa

Rica (primary) and El Salvador) that collect and process flash flood-related forecast data and issue timely products to NMHS of Belize, Guatemala, Honduras, Nicaragua, and Panama, which can refine forecasts with local knowledge and observations and then disseminate this information further to relevant disaster preparedness and response agencies (UCAR 2010; WMO 2017; Barrett et al. 2018).

4.5 | Participatory Science

Participatory science refers to activities or research, “in which the general public contributes to the production of scientific knowledge, either alone, or more often in collaboration with professional scientists and scientific institutions” (Tan et al. 2022, 3). Integrating co-creation, co-production, and co-implementation, participatory science projects allow end-users to make FEWS that are more accessible, becoming more than just social sensors (via mobile applications) and actually gathering and sharing data, and quality checking and verifying decentralized communication on social media in real-time (Tan et al. 2022). The benefits of participatory science are that communication becomes two-way, and citizens can become interested, engaged, and supportive of science, making them warning-ready citizens. A combination of participatory science, citizen observatories, and hydrological modeling can reduce flood damages by 45% (Tan et al. 2022). For example, nearly all of the 1200 environmental sensors that are part of the Early Warning System of Medellín and the Aburrá Valley in Colombia are managed by citizens (Palmer 2023), and the system includes a Citizen Scientist Seedbeds program which educates youth on scientific topics and allows them to take part in related projects to better understand local hazards and risks (SIATA 2022).

4.6 | Artificial Intelligence and Related Advanced Technology

Considering access to, reliance on, and capabilities of modern technology has become integral to an effective people-centered FEWS (e.g., replacing hydrological and hydraulic processes with Rain-on-Grid modeling for improved flood risk management, studied in Chaitanya et al. 2024). Emerging technologies include artificial intelligence (AI), which refers to “a wide range of technologies that enable machines to perform tasks typically requiring human intelligence, such as reasoning, learning, and self-correction” (WMO 2024, 17), as well as knowledge representation, planning, natural language processing, and moving and manipulating objects (Lamsal and Kumar 2020).

AI can enhance disaster risk knowledge by assessing vulnerability and exposure data, and it can also speed up hazard detection and monitoring by enhancing predictive analytics and real-time data assessments, and accelerate risk communication and warning dissemination by optimizing which channels information should be sent, to whom, and in what languages (UNU-EHS 2024). Human oversight should ensure AI implementation in FEWS addresses the digital divide, avoids biases, and adheres to guidelines for ethical use (UNU-EHS 2024).

Google's Flood Hub application, available in over 100 countries as of early 2025, is entirely AI-run via machine learning; a hydrology and inundation model is trained using publicly available data to produce accurate and timely forecasts and alerts, and where available, flood inundation maps. Users are able to provide streamflow data via an open-source Caravan project to expand the reach and capabilities of the system, and Google works with local organizations to understand how Flood Hub can support community preparedness and response (Google Research 2025).

4.7 | Bottom-Up

FEWS should be planned and integrated into the community through a bottom-up approach in which end-users are the central consideration (Kelman and Glantz 2014). A strong reliance on flood authorities may weaken individual motivation to take action against flooding (Maidl and Buchecker 2015). Participation from end-users, in addition to that of flood authorities, allows the flood risk perspectives of both parties to be heard and unified, which will then create local capacity to better understand and implement preparatory and protective actions against flood events (Rollason et al. 2018). End-users may also feel more empowered to address flooding, make appropriate and adequate decisions, and have greater trust in the FEWS. An example of a bottom-up approach to FEWS is the CBFWS (in pilot testing as of 2025) in several communities in the Hindu Kush Himalayan region; the community establishes, manages, and operates the system, and the CBFWS are designed so that the community is involved in all four steps of the flood early warning value chain (Shakya et al. 2019).

One way to include community and stakeholder considerations in the design and operations of a FEWS is to involve social science research in the development of the forecast products, using findings from focus groups, surveys, and interviews with involved communities to inform the design and dissemination of such information. For example, the Nurture Nature Center research team collaborated with the US NWS WPC on a social science research study to explore the needs for urban rainfall rates in communities across the country to inform the development of a new flood forecast product (NA23OAR4590137-T1-01). Getting user input early on in the process ensures the product is useful for decision-making and will be integrated in flood early warning systems.

5 | Discussion

Our analysis of the “FEWS Around the World” repository shows that most of the systems reviewed have key characteristics of an effective FEWS outlined in Kelman and Glantz (2014) over a decade ago, but fewer have additional characteristics recommended in this paper (see Figure 2) to account for recent technological evolution and global wicked problems like climate change. Ensuring a FEWS is designed with all 14 key characteristics is important to support effectiveness; together, they can help a system overcome identified barriers in each step of the flood early warning value chain, and new and future barriers. For example, securing and maintaining a *preparedness fund* bolsters system

operation against changes in political or budgetary support, and opportunities for *participatory science* represent one way to implement FEWS through a *bottom-up* approach and also increase *human capacity* and further *integrate* the system into the areas it is operational. Several systems in the repository are people-centric, focused on empowering the communities they serve (e.g., CBFWS from Practical Action and ICIMOD, FijiFFGS in Fiji, and Google's Flood Hub), building resilience (e.g., FloodNet NYC and Smart FEWS in the Cayman Islands), or developing preparedness and response capabilities (e.g., Flood-PROOFS in the Greater Horn of Africa and FEWS in Flanders, Belgium), or all three (e.g., SUFAL in northern Bangladesh).

In our literature and key characteristics reviews, we find that disaster risk and hazard knowledge form the basis of a FEWS (step 1), there is a growing trend to transition from hazard- to impact-based monitoring and warning (step 2), and that risk communication and message dissemination are both integral to an effective warning system (step 3). Sukhwani et al. (2019) study several barriers in these steps in operational FEWS that arose during disasters in Cameron Highlands, Malaysia (October 2013), North Kyushu, Japan (July 2017), and Sri Lanka (May 2017); they support our key characteristics of an effective FEWS by placing emphasis on *timely warnings* and *flexible/adaptable* EWS, and discuss accessible and *impact-based warnings*, *governance structures* with clearly defined roles, *partnerships* with effective and robust communication, and *technological advancements* in forecasting.

Previous studies on disaster risk reduction have also explored the benefit or illustrate the need for community-based FEWS. For example, Baudoin et al. (2016) study community participation in top-down (Kenya), bottom-up (Sri Lanka), and hybrid (Hawai'i, United States) approaches to EWS and suggest that system design and implementation should move from being expert-driven to community-driven, also reiterating the importance of considering all vulnerable populations. Samansiri et al. (2023) complete a literature review and interpretive structural modeling to identify and better understand critical failure factors in FEWS; they share the importance of community engagement in designing people-centric warning systems, without which several barriers arise. We contribute to these studies by exploring how community engagement might be realized or achieved in each step of the flood early warning value chain and focus on designing people-centered FEWS that combat identified barriers and critical failure factors.

People-centered key characteristics (e.g., *integration*, *impact-based warnings*, and *bottom-up*) highlight that communities should not solely rely on physical flood protection. Wang et al. (2023) study the correlation between human-flood distancing and reducing flood fatalities and displacement, also considering the influence of flood protection level (FPL) and the decision to relocate. They find that people in areas with higher FPL are less likely to relocate, suggesting local trust in flood protection strategies. While physical flood protection and distancing may reduce risk in some situations, underestimating risk and not having a FEWS can still lead to flood impacts. Rufat et al. (2015) complete a meta-analysis of flood disaster case studies and find examples of social vulnerability in Europe where structural protection and institutional management caused a decrease in flood

awareness and adoption of self-protective skills. While there might be trade-offs between structural protection and people-centered approaches, including physical and social approaches is important, as found by Mehryar and Surminski (2022) where fuzzy cognitive mapping with a flood resilience measurement framework found that increasing flood risk and management knowledge and awareness is perceived as particularly impactful for building community resilience.

With the evolving nature of flooding, simply having a system in place is not enough without considering the characteristics of that system and how it can support and be adaptable to community needs and a changing climate. For example, previous literature has explored the marginalization of Indigenous knowledge in favor of scientific knowledge (e.g., Balay-As et al. 2018) and the political economy of fostering resilience through CBFWEWS (Gladfelter 2018). We recognize the complexity of data sharing and colonial legacies of early warning dissemination, as well as how age, gender, race, disability, sexuality, and other demographics can factor into vulnerability and flood resiliency. Future research should review intersectional studies on these topics and (flood) disasters in the context of our proposed key characteristics of an effective FEWS. Additionally, further research on what promotes flood risk perceptions and decisions to engage in protective behaviors will support FEWS design, implementation, and effectiveness. There is a lack of longitudinal studies that explore changes in risk perception and engaging in protective behaviors over time, and similarly with impact-based warnings, whether levels of risk perceptions and personal thresholds that dictate taking protective behaviors change between different hazards or threats (Weyrich, Scolobig, et al. 2020).

6 | Conclusions

Historically, flood disaster risk governance entailed providing monetary support to affected communities to help them rebuild, with very little investment in adaptation or preventive activities. Today, flood disaster risk governance requires all relevant stakeholders to understand and assess risk exposure and dictate who is responsible for managing it; efforts backed by legislation, funding, and management and enforcement from a governmental entity (Golding 2022). The number of flood disasters and related mortalities has been on a downward trend since 2000 (Perera et al. 2020); this can be attributed to continued investment in flood mitigation and risk reduction strategies, which include FEWS. With effective FEWS, greater emphasis is placed on enhancing disaster risk and hazard knowledge, increasing flood preparedness and response capabilities, and helping communities become more resilient (Golding 2022). For example, two-thirds of respondents in Practical Action's assessment of the Lower Karnali River Basin said their disaster risk knowledge and understanding improved over the past 5 years due to early warning system projects (Practical Action 2023). To build on this progress, FEWS need to be accessible and actionable, tailored to the needs of the community and prioritize co-creation, co-production, and co-implementation with relevant stakeholders. Our review suggests that newer FEWS and initiatives like EW4ALL are making the inclusion of people and society in all

aspects of early warnings a core value; it is imperative this trend continues.

Illustrated in the “FEWS Around the World” repository, the design of FEWS varies globally, but effectiveness is improved with certain key characteristics, including *continuity, timeliness, transparency, flexibility/adaptability, integration, human capacity, catalysts/patterns, impact-based warnings, partnerships, preparedness funding, modern technologies, governance structures*, and implemented through *bottom-up* approaches. However, challenges remain because flood impacts have become more complex and outside the normal due to climate change, which complicates the effectiveness of systems. Further, rapidly advancing technology and the expansion of data available require resources and consideration of equity in use.

While over 50% of countries have now implemented multi-hazard early warning systems (WMO 2024), and this paper posits several key characteristics of an effective FEWS that aim to address these challenges, there are many questions left to address: How should the impacts of climate change and technology be understood in the context of FEWS? How might FEWS evolve in the future by exploring concepts such as resilience and transilience frameworks? We consider these questions in our next phase of research.

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Data Availability Statement

The data that support the findings of this study are openly available in the public domain as this was a literature review article. References to these sources are included in the paper. A repository of FEWS analyzed is available as a spreadsheet at <https://tinyurl.com/FEWSaroundtheworld>.

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Appendix A

TABLE A1 | Examples of flood early warning systems.

England see Mehring et al. (2018)	<i>Current Situation:</i> - Flood management in England was tied to agricultural needs until the late 1900s, following several floods. Since then, it has involved creating flood plans, and a focus on becoming more flood resistant and resilient. <i>Challenge:</i> - A priority of flood management is to integrate flood vulnerable communities in decision-making, but it has been difficult moving from historically technocratic and top-down ways of flood management, to a more inclusive and democratized approach. <i>Solution:</i> - Mehring et al. (2018) complete social science research to determine what a 'good' approach to bottom-up flood management might be. - The study finds that <i>good</i> engagement is dependent on the social construction of the community in question but authorities should engage in two-way partnerships with residents and, "...aim to use social learning to co-create knowledge, enhance social capital and increase the resilience of society towards flooding" (p. 111).
Uganda see Mitheu et al. (2023)	<i>Current Situation:</i> - Communities in the Katakwi District of Uganda benefit from hazard-based warnings, supported by forecast information from international databases and organizations. <i>Challenge:</i> - The current hazard-based flood-early warning system (HbFEWStS) has difficulty in providing useful impact-based information for local farmers to complete preparatory actions. <i>Solution:</i> - Mitheu et al. (2023) study the effectiveness of incorporating crop cycles and crop-impact information into forecasts that trigger warning thresholds, thus creating an impact-based flood early-warning trigger system (IbFEWStS). - The study finds that trigger thresholds can be lowered during harvesting and planting season to avoid crop damages, and historical flood-impact data on crops can be used to determine if a warning should be released, even if the forecast does not meet the normal trigger threshold. This IbFEWStS better detects impactful flood events and enhances risk perceptions which convince people to take protective actions.
Spain see Cecilia et al. (2021)	<i>Current Situation:</i> - Climate change is a wicked problem that has magnified the impacts of natural disasters, like flooding. Understanding the importance of early detection to save lives, it has become paramount to find new ways to effectively communicate on disasters. <i>Challenge:</i> - Various research has studied ways that social media, among other forms of communication platforms, can be used to improve warning systems. However, research has only focused on post-event analysis or the information is not actionable, among other noted limitations. <i>Solution:</i> - Cecilia et al. (2021) tested <i>WATERSensing</i>, a social sensing application of the Spanish government funded project WATERoT, which processes Twitter (now X), RSS feeds, and Telegram messages/opinions. - Using Storm Gloria (2020) as a case study, it is found that "the perception of the problem in social networks and the media perfectly correlates with the evolution of the natural disaster" (p. 94) and means that opinions on social media can help professional stakeholders in their decision-making.

(Continues)

TABLE A1 | (Continued)

Philippines see Balay-As et al. (2018)	<i>Current Situation:</i> • The indigenous Kankanaey of northern Philippines have increasingly integrated scientific knowledge into indigenous knowledge to better mitigate disasters and increase indigenous people's resilience to natural hazards. <i>Challenge:</i> • Scientific research fails to recognize that for centuries, indigenous peoples have successfully responded to natural hazards using their own methods. • Indigenous peoples often do not fully understand the science of climate change, but they have found that historically successful mitigative practices may not be as effective anymore. • Incorporating indigenous knowledge into government funded disaster risk reduction (DRR) is difficult, and outside help/technologically-based EWS is often associated with hidden agendas (thrust onto indigenous peoples), social and economic injustices, and commercialization, all of which erodes indigenous knowledge. <i>Solution:</i> • Balay-As et al. (2018) find that to better incorporate indigenous knowledge into FEWS, "a participatory framework by which indigenous peoples can actively engage and address the power relations and injustices that impede the meaningful integration of indigenous and scientific knowledge in DRR" (p. 22) is crucial.
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